

Step-by-Step Solutions

Exercise 13B: Operations on Algebraic Expressions

Topic Overview: This exercise focuses on the fundamental arithmetic operations of **addition** and **subtraction** on algebraic expressions and polynomials. The core mathematical principle applied throughout these solutions is the **combining of like terms** (terms having identical literal coefficients). Under addition, numerical coefficients of like terms are summed; under subtraction, the sign of each term of the polynomial to be subtracted is inverted before addition.

Question 1: Add the following expressions

(i) $2x^2, -5x^2, -x^2, 6x^2$

Solution:

Combine all terms as they are like terms (literal coefficient is x^2):

$$\begin{aligned}\text{Sum} &= 2x^2 + (-5x^2) + (-x^2) + 6x^2 \\ &= (2 - 5 - 1 + 6)x^2 \\ &= 2x^2\end{aligned}$$

(ii) $x^2 - 2xy + 3y^2, 5y^2 + 3xy - 6x^2$

Solution:

Group the like terms (by x^2 , xy , and y^2):

$$\begin{aligned}\text{Sum} &= (x^2 - 2xy + 3y^2) + (-6x^2 + 3xy + 5y^2) \\ &= (x^2 - 6x^2) + (-2xy + 3xy) + (3y^2 + 5y^2) \\ &= -5x^2 + xy + 8y^2\end{aligned}$$

(iii) $2x + 9y - 7z, 3y + z - 3x, 2z - 4y - x$

Solution:

Group the variables x , y , and z :

$$\begin{aligned}\text{Sum} &= (2x + 9y - 7z) + (-3x + 3y + z) + (-x - 4y + 2z) \\ &= (2x - 3x - x) + (9y + 3y - 4y) + (-7z + z + 2z) \\ &= -2x + 8y - 4z\end{aligned}$$

(iv) $2ab + 3bc - 5ca, 4bc - 3ab + 7ca, 2ca - ab - 5bc$

Solution:

Group the literal components ab , bc , and ca :

$$\begin{aligned}\text{Sum} &= (2ab + 3bc - 5ca) + (-3ab + 4bc + 7ca) + (-ab - 5bc + 2ca) \\ &= (2ab - 3ab - ab) + (3bc + 4bc - 5bc) + (-5ca + 7ca + 2ca) \\ &= -2ab + 2bc + 4ca\end{aligned}$$

(v) $3x^3 + 2x^2 - 6x + 3$, $2x^3 - 3x^2 - x - 4$, $1 + 2x - 3x^2 - 4x^3$

Solution:

Arrange and group terms of the same degree (from x^3 down to constants):

$$\begin{aligned}\text{Sum} &= (3x^3 + 2x^2 - 6x + 3) + (2x^3 - 3x^2 - x - 4) + (-4x^3 - 3x^2 + 2x + 1) \\ &= (3x^3 + 2x^3 - 4x^3) + (2x^2 - 3x^2 - 3x^2) + (-6x - x + 2x) + (3 - 4 + 1) \\ &= (1)x^3 + (-4)x^2 + (-5)x + (0) \\ &= x^3 - 4x^2 - 5x\end{aligned}$$

(vi) $3n^2 + 5mn - 6m^2$, $2m^2 - 3mn - 4n^2$, $2mn - 3m^2 - 7n^2$

Solution:

Group the terms containing m^2 , mn , and n^2 :

$$\begin{aligned}\text{Sum} &= (-6m^2 + 5mn + 3n^2) + (2m^2 - 3mn - 4n^2) + (-3m^2 + 2mn - 7n^2) \\ &= (-6m^2 + 2m^2 - 3m^2) + (5mn - 3mn + 2mn) + (3n^2 - 4n^2 - 7n^2) \\ &= -7m^2 + 4mn - 8n^2\end{aligned}$$

(vii) $3z^3 - z^2 + 5$, $1 - 2z + 2z^2$, $3 + 2z - z^3$

Solution:

Group the terms by powers of z :

$$\begin{aligned}\text{Sum} &= (3z^3 - z^2 + 5) + (2z^2 - 2z + 1) + (-z^3 + 2z + 3) \\ &= (3z^3 - z^3) + (-z^2 + 2z^2) + (-2z + 2z) + (5 + 1 + 3) \\ &= 2z^3 + z^2 + 0z + 9 \\ &= 2z^3 + z^2 + 9\end{aligned}$$

Question 2: Simplify

(i) $5x + 3y - 8z + 2y - 3x + 5z + z - 7y - 2x$

Solution:Group the like terms for x , y , and z :

$$\begin{aligned}\text{Expression} &= (5x - 3x - 2x) + (3y + 2y - 7y) + (-8z + 5z + z) \\ &= (0)x + (-2)y + (-2)z \\ &= -2y - 2z\end{aligned}$$

(ii) $4x^3 - 2x^2 + 5x - 1 + 8x + x^2 - 6x^3 + 7 - 6x + 3 - 3x^2 - x^3$

Solution:

Group terms of identical degrees:

$$\begin{aligned}\text{Expression} &= (4x^3 - 6x^3 - x^3) + (-2x^2 + x^2 - 3x^2) + (5x + 8x - 6x) + (-1 + 7 + 3) \\ &= -3x^3 - 4x^2 + 7x + 9\end{aligned}$$

(iii) $2x^2 - 3xy^2 + y^2 + x^2 - xy^2 + y^2$

Solution:Group like terms (x^2 , xy^2 , and y^2):

$$\begin{aligned}\text{Expression} &= (2x^2 + x^2) + (-3xy^2 - xy^2) + (y^2 + y^2) \\ &= 3x^2 - 4xy^2 + 2y^2\end{aligned}$$

(iv) $2 - 3z^2 + 5yz + 7y^2 - 8 + z^2 - 6yz - 9y^2 + 1 - 2z^2 - 2yz - y^2$

Solution:Group terms by variable type (y^2 , yz , z^2 , and constants):

$$\begin{aligned}\text{Expression} &= (7y^2 - 9y^2 - y^2) + (5yz - 6yz - 2yz) + (-3z^2 + z^2 - 2z^2) + (2 - 8 + 1) \\ &= -3y^2 - 3yz - 4z^2 - 5\end{aligned}$$

(v) $2m - 3n + 5p + 2m + n - 2p - 3m - 4n + p$

Solution:Group variables m , n , and p :

$$\begin{aligned}\text{Expression} &= (2m + 2m - 3m) + (-3n + n - 4n) + (5p - 2p + p) \\ &= m - 6n + 4p\end{aligned}$$

Question 3: Adjacent sides of a rectangle are $3a - b$ and $6b - a$. Find its perimeter.

Solution:

The perimeter (P) of a rectangle is given by the formula:

$$P = 2 \times (\text{Length} + \text{Breadth})$$

Given that the adjacent sides are:

$$\text{Length} = 3a - b$$

$$\text{Breadth} = 6b - a$$

First, find the sum of the adjacent sides (Length + Breadth):

$$\text{Sum} = (3a - b) + (6b - a) = (3a - a) + (-b + 6b) = 2a + 5b$$

Now, multiply the sum by 2:

$$P = 2(2a + 5b) = 4a + 10b$$

Answer: The perimeter of the rectangle is $4a + 10b$ units.

Question 4: Find the perimeter of a triangle whose sides are $2y + 3z$, $z - y$, $4y - 2z$.

Solution:

The perimeter of a triangle is the sum of the lengths of all three sides:

$$\text{Perimeter} = \text{Side}_1 + \text{Side}_2 + \text{Side}_3$$

Given sides:

$$\text{Side}_1 = 2y + 3z$$

$$\text{Side}_2 = z - y = -y + z$$

$$\text{Side}_3 = 4y - 2z$$

Adding the three expressions together:

$$\begin{aligned}\text{Perimeter} &= (2y + 3z) + (-y + z) + (4y - 2z) \\ &= (2y - y + 4y) + (3z + z - 2z) \\ &= 5y + 2z\end{aligned}$$

Answer: The perimeter of the triangle is $5y + 2z$ units.

Question 5: Subtract

- (i)
- $3a - 2b + 4c$
- from
- $5a - 3b - 5c$

Solution:

Subtract the first expression from the second:

$$\begin{aligned}
 \text{Difference} &= (5a - 3b - 5c) - (3a - 2b + 4c) \\
 &= 5a - 3b - 5c - 3a + 2b - 4c \\
 &= (5a - 3a) + (-3b + 2b) + (-5c - 4c) \\
 &= \mathbf{2a - b - 9c}
 \end{aligned}$$

- (ii)
- $5x^2 - 3xy - 7y^2$
- from
- $3x^2 - xy - 2y^2$

Solution:

Subtract the first expression from the second:

$$\begin{aligned}
 \text{Difference} &= (3x^2 - xy - 2y^2) - (5x^2 - 3xy - 7y^2) \\
 &= 3x^2 - xy - 2y^2 - 5x^2 + 3xy + 7y^2 \\
 &= (3x^2 - 5x^2) + (-xy + 3xy) + (-2y^2 + 7y^2) \\
 &= \mathbf{-2x^2 + 2xy + 5y^2}
 \end{aligned}$$

- (iii)
- $3p^3 - 5p^2q + 2q^2$
- from
- $q^3 + p^2q + p^3 - 4p^3$

Solution:

First, simplify the second expression:

$$q^3 + p^2q + p^3 - 4p^3 = q^3 + p^2q - 3p^3$$

Now subtract the first expression from it:

$$\begin{aligned}
 \text{Difference} &= (q^3 + p^2q - 3p^3) - (3p^3 - 5p^2q + 2q^2) \\
 &= q^3 + p^2q - 3p^3 - 3p^3 + 5p^2q - 2q^2 \\
 &= (-3p^3 - 3p^3) + (p^2q + 5p^2q) - 2q^2 + q^3 \\
 &= \mathbf{-6p^3 + 6p^2q - 2q^2 + q^3}
 \end{aligned}$$

- (iv)
- $ab - bc - ca$
- from
- $3ab + 2bc - 4ca$

Solution:

Subtract the first expression from the second:

$$\begin{aligned}
 \text{Difference} &= (3ab + 2bc - 4ca) - (ab - bc - ca) \\
 &= 3ab + 2bc - 4ca - ab + bc + ca \\
 &= (3ab - ab) + (2bc + bc) + (-4ca + ca) \\
 &= \mathbf{2ab + 3bc - 3ca}
 \end{aligned}$$

- (v)
- $3z^3 - 2z^2 + 7z - 8$
- from
- $8 - z - z^2$

Solution:

Subtract the first expression from the second:

$$\begin{aligned}\text{Difference} &= (8 - z - z^2) - (3z^3 - 2z^2 + 7z - 8) \\ &= 8 - z - z^2 - 3z^3 + 2z^2 - 7z + 8 \\ &= -3z^3 + (-z^2 + 2z^2) + (-z - 7z) + (8 + 8) \\ &= -3z^3 + z^2 - 8z + 16\end{aligned}$$

(vi) $2abc - a^2 - b^2$ from $b^2 + a^2 - 2abc$

Solution:

Subtract the first expression from the second:

$$\begin{aligned}\text{Difference} &= (b^2 + a^2 - 2abc) - (2abc - a^2 - b^2) \\ &= b^2 + a^2 - 2abc - 2abc + a^2 + b^2 \\ &= (a^2 + a^2) + (b^2 + b^2) + (-2abc - 2abc) \\ &= 2a^2 + 2b^2 - 4abc\end{aligned}$$

Question 6

- (i) Subtract $6x^3 - 5x^2 + 4x - 3$ from the sum of $x + 2x^2 - 3x^3$ and $2 - x^2 + 6x - x^3$.

Solution:

Step 1: Calculate the sum of the two specified expressions:

$$\begin{aligned}\text{Sum} &= (x + 2x^2 - 3x^3) + (2 - x^2 + 6x - x^3) \\ &= (-3x^3 - x^3) + (2x^2 - x^2) + (x + 6x) + 2 \\ &= -4x^3 + x^2 + 7x + 2\end{aligned}$$

Step 2: Subtract the target expression $6x^3 - 5x^2 + 4x - 3$ from the obtained sum:

$$\begin{aligned}\text{Result} &= (-4x^3 + x^2 + 7x + 2) - (6x^3 - 5x^2 + 4x - 3) \\ &= -4x^3 + x^2 + 7x + 2 - 6x^3 + 5x^2 - 4x + 3 \\ &= (-4x^3 - 6x^3) + (x^2 + 5x^2) + (7x - 4x) + (2 + 3) \\ &= -10x^3 + 6x^2 + 3x + 5\end{aligned}$$

- (ii) Subtract the sum of $a + 2b - 3c$ and $2c - 3b - 4a$ from the sum of $5b - 4c + a$ and $2c - 3b - 4a$.

Solution (Algebraic Reduction Method):

Let us define:

$$\begin{aligned}P &= a + 2b - 3c \\ Q &= 2c - 3b - 4a \\ R &= 5b - 4c + a\end{aligned}$$

We are asked to subtract the first sum ($P + Q$) from the second sum ($R + Q$):

$$\begin{aligned}\text{Required Expression} &= (R + Q) - (P + Q) \\ &= R + Q - P - Q \\ &= R - P\end{aligned}$$

Now, substitute the expressions for R and P :

$$\begin{aligned}R - P &= (5b - 4c + a) - (a + 2b - 3c) \\ &= a + 5b - 4c - a - 2b + 3c \\ &= (a - a) + (5b - 2b) + (-4c + 3c) \\ &= 3b - c\end{aligned}$$

We can also verify this by expanding both sums fully:

$$\text{Sum 1 } (P + Q) = (a - 4a) + (2b - 3b) + (-3c + 2c) = -3a - b - c$$

$$\text{Sum 2 } (R + Q) = (a - 4a) + (5b - 3b) + (-4c + 2c) = -3a + 2b - 2c$$

$$\text{Difference} = (-3a + 2b - 2c) - (-3a - b - c) = -3a + 2b - 2c + 3a + b + c = 3b - c$$

Both methods yield the identical elegant result: $3b - c$.

- (iii) Subtract the sum of $x^2 - 5xy + 2y^2$ and $y^2 - 2xy - 3x^2$ from the sum of $6x^2 - 8xy - y^2$ and $2xy - 2y^2 - x^2$.

Solution:

Step 1: Find the first sum (Sum_1):

$$\begin{aligned}\text{Sum}_1 &= (x^2 - 5xy + 2y^2) + (-3x^2 - 2xy + y^2) \\ &= (x^2 - 3x^2) + (-5xy - 2xy) + (2y^2 + y^2) \\ &= -2x^2 - 7xy + 3y^2\end{aligned}$$

Step 2: Find the second sum (Sum_2):

$$\begin{aligned}\text{Sum}_2 &= (6x^2 - 8xy - y^2) + (-x^2 + 2xy - 2y^2) \\ &= (6x^2 - x^2) + (-8xy + 2xy) + (-y^2 - 2y^2) \\ &= 5x^2 - 6xy - 3y^2\end{aligned}$$

Step 3: Subtract Sum_1 from Sum_2 :

$$\begin{aligned}\text{Difference} &= \text{Sum}_2 - \text{Sum}_1 \\ &= (5x^2 - 6xy - 3y^2) - (-2x^2 - 7xy + 3y^2) \\ &= 5x^2 - 6xy - 3y^2 + 2x^2 + 7xy - 3y^2 \\ &= (5x^2 + 2x^2) + (-6xy + 7xy) + (-3y^2 - 3y^2) \\ &= 7x^2 + xy - 6y^2\end{aligned}$$

Question 7

(i) What should be subtracted from $x + 2y - 3z$ to get $3x - 2y + z$?

Solution:

Let the required algebraic expression be P . Then, according to the problem:

$$\begin{aligned}(x + 2y - 3z) - P &= 3x - 2y + z \\ P &= (x + 2y - 3z) - (3x - 2y + z) \\ &= x + 2y - 3z - 3x + 2y - z \\ &= (x - 3x) + (2y + 2y) + (-3z - z) \\ &= -2x + 4y - 4z\end{aligned}$$

(ii) What should be subtracted from $2x^2 - y^2 + 4z^2$ to get $x^2 + y^2 - z^2$?

Solution:

Let the required algebraic expression be P . Then:

$$\begin{aligned}(2x^2 - y^2 + 4z^2) - P &= x^2 + y^2 - z^2 \\ P &= (2x^2 - y^2 + 4z^2) - (x^2 + y^2 - z^2) \\ &= 2x^2 - y^2 + 4z^2 - x^2 - y^2 + z^2 \\ &= (2x^2 - x^2) + (-y^2 - y^2) + (4z^2 + z^2) \\ &= x^2 - 2y^2 + 5z^2\end{aligned}$$

(iii) What should be subtracted from $1 + x - x^2$ to get $2x + x^2$?

Solution:

Let the required algebraic expression be P . Then:

$$\begin{aligned}(1 + x - x^2) - P &= 2x + x^2 \\ P &= (1 + x - x^2) - (2x + x^2) \\ &= 1 + x - x^2 - 2x - x^2 \\ &= 1 + (x - 2x) + (-x^2 - x^2) \\ &= 1 - x - 2x^2 \quad (\text{or } -2x^2 - x + 1)\end{aligned}$$

Question 8

(i) What should be added to $7a - 9b + 13c$ to get $9a + b - c$?

Solution:

Let the required algebraic expression be Q . Then, according to the problem:

$$\begin{aligned}
 (7a - 9b + 13c) + Q &= 9a + b - c \\
 Q &= (9a + b - c) - (7a - 9b + 13c) \\
 &= 9a + b - c - 7a + 9b - 13c \\
 &= (9a - 7a) + (b + 9b) + (-c - 13c) \\
 &= \mathbf{2a + 10b - 14c}
 \end{aligned}$$

(ii) What should be added to $1 + 2x - 3x^2$ to get $x^2 - x - 1$?

Solution:

Let the required algebraic expression be Q . Then:

$$\begin{aligned}
 (1 + 2x - 3x^2) + Q &= x^2 - x - 1 \\
 Q &= (x^2 - x - 1) - (1 + 2x - 3x^2) \\
 &= x^2 - x - 1 - 1 - 2x + 3x^2 \\
 &= (x^2 + 3x^2) + (-x - 2x) + (-1 - 1) \\
 &= \mathbf{4x^2 - 3x - 2}
 \end{aligned}$$

(iii) What should be added to $m^2 - 2mn + 5n^2$ to get $n^2 + mn - m^2$?

Solution:

Let the required algebraic expression be Q . Then:

$$\begin{aligned}
 (m^2 - 2mn + 5n^2) + Q &= n^2 + mn - m^2 \\
 Q &= (n^2 + mn - m^2) - (m^2 - 2mn + 5n^2) \\
 &= n^2 + mn - m^2 - m^2 + 2mn - 5n^2 \\
 &= (-m^2 - m^2) + (mn + 2mn) + (n^2 - 5n^2) \\
 &= \mathbf{-2m^2 + 3mn - 4n^2}
 \end{aligned}$$

Question 9: Find the excess of $4p^2 - 2pq + 3q^2$ over $2p^2 - pq + 4q^2$.

Solution:

To find the excess of one expression over another, subtract the second expression from the first:

$$\begin{aligned}\text{Excess} &= (4p^2 - 2pq + 3q^2) - (2p^2 - pq + 4q^2) \\ &= 4p^2 - 2pq + 3q^2 - 2p^2 + pq - 4q^2 \\ &= (4p^2 - 2p^2) + (-2pq + pq) + (3q^2 - 4q^2) \\ &= 2p^2 - pq - q^2\end{aligned}$$

Answer: The excess is $2p^2 - pq - q^2$.

Question 10: By how much does $3x^3 - 5x^2 + 2x - 3$ exceed $2x^3 - 3x^2 + x + 1$?

Solution:

To find by how much an expression exceeds another, compute their difference:

$$\begin{aligned}\text{Difference} &= (3x^3 - 5x^2 + 2x - 3) - (2x^3 - 3x^2 + x + 1) \\ &= 3x^3 - 5x^2 + 2x - 3 - 2x^3 + 3x^2 - x - 1 \\ &= (3x^3 - 2x^3) + (-5x^2 + 3x^2) + (2x - x) + (-3 - 1) \\ &= x^3 - 2x^2 + x - 4\end{aligned}$$

Answer: It exceeds by $x^3 - 2x^2 + x - 4$.

Question 11: How much is $-x^2 + 7y^2 - 3xy$ less than $2x^2 - y^2 + xy$?

Solution:

To find how much an expression is less than another, subtract the first expression from the second:

$$\begin{aligned}\text{Difference} &= (2x^2 - y^2 + xy) - (-x^2 + 7y^2 - 3xy) \\ &= 2x^2 - y^2 + xy + x^2 - 7y^2 + 3xy \\ &= (2x^2 + x^2) + (xy + 3xy) + (-y^2 - 7y^2) \\ &= 3x^2 + 4xy - 8y^2\end{aligned}$$

Answer: It is less by $3x^2 + 4xy - 8y^2$.

Question 12: How much is $x^3 - 3x^2 + 5x - 2$ less than $3 - 2x + x^2 - x^3$?

Solution:

Subtract the first expression from the second:

$$\begin{aligned}\text{Difference} &= (3 - 2x + x^2 - x^3) - (x^3 - 3x^2 + 5x - 2) \\ &= 3 - 2x + x^2 - x^3 - x^3 + 3x^2 - 5x + 2 \\ &= (-x^3 - x^3) + (x^2 + 3x^2) + (-2x - 5x) + (3 + 2) \\ &= -2x^3 + 4x^2 - 7x + 5\end{aligned}$$

Answer: It is less by $-2x^3 + 4x^2 - 7x + 5$.

Question 13: If $x = 2a^2 + 3b^2 - 5ab$, $y = b^2 - 3a^2 + 7ab$ and $z = 6a^2 - b^2 + ab$, find:

(i) $x + y - z$

Solution:

First, find the sum $x + y$:

$$\begin{aligned}x + y &= (2a^2 + 3b^2 - 5ab) + (-3a^2 + b^2 + 7ab) \\&= (2a^2 - 3a^2) + (3b^2 + b^2) + (-5ab + 7ab) \\&= -a^2 + 4b^2 + 2ab\end{aligned}$$

Now, subtract $z = 6a^2 - b^2 + ab$ from the sum:

$$\begin{aligned}x + y - z &= (-a^2 + 4b^2 + 2ab) - (6a^2 - b^2 + ab) \\&= -a^2 + 4b^2 + 2ab - 6a^2 + b^2 - ab \\&= (-a^2 - 6a^2) + (4b^2 + b^2) + (2ab - ab) \\&= -7a^2 + 5b^2 + ab\end{aligned}$$

(ii) $x - y + z$

Solution:

First, find the difference $x - y$:

$$\begin{aligned}x - y &= (2a^2 + 3b^2 - 5ab) - (-3a^2 + b^2 + 7ab) \\&= 2a^2 + 3b^2 - 5ab + 3a^2 - b^2 - 7ab \\&= (2a^2 + 3a^2) + (3b^2 - b^2) + (-5ab - 7ab) \\&= 5a^2 + 2b^2 - 12ab\end{aligned}$$

Now, add $z = 6a^2 - b^2 + ab$ to this difference:

$$\begin{aligned}x - y + z &= (5a^2 + 2b^2 - 12ab) + (6a^2 - b^2 + ab) \\&= (5a^2 + 6a^2) + (2b^2 - b^2) + (-12ab + ab) \\&= 11a^2 + b^2 - 11ab\end{aligned}$$

Question 14: The perimeter of a triangle is $8 + 13a + 7a^2$ and two of its sides are $2a^2 + 3a + 2$ and $3a^2 - 4a - 1$. Find the third side.

Solution:

The perimeter (P) of a triangle with sides s_1 , s_2 , and s_3 is given by:

$$P = s_1 + s_2 + s_3 \implies s_3 = P - (s_1 + s_2)$$

Given values:

$$P = 7a^2 + 13a + 8$$

$$s_1 = 2a^2 + 3a + 2$$

$$s_2 = 3a^2 - 4a - 1$$

First, compute the sum of the two given sides ($s_1 + s_2$):

$$s_1 + s_2 = (2a^2 + 3a + 2) + (3a^2 - 4a - 1) = (2a^2 + 3a^2) + (3a - 4a) + (2 - 1) = 5a^2 - a + 1$$

Now, subtract this sum from the perimeter to find the third side (s_3):

$$\begin{aligned} s_3 &= (7a^2 + 13a + 8) - (5a^2 - a + 1) \\ &= 7a^2 + 13a + 8 - 5a^2 + a - 1 \\ &= (7a^2 - 5a^2) + (13a + a) + (8 - 1) \\ &= 2a^2 + 14a + 7 \end{aligned}$$

Answer: The third side of the triangle is $2a^2 + 14a + 7$ units.

Question 15: The perimeter of a rectangle is $16x^3 - 6x^2 + 12x + 4$. If one of its sides is $8x^2 + 3x$, find the other side.

Solution:

The perimeter (P) of a rectangle is defined as:

$$P = 2 \times (\text{Side}_1 + \text{Side}_2)$$

Therefore, the sum of the two adjacent sides is half of the perimeter:

$$\text{Side}_1 + \text{Side}_2 = \frac{P}{2}$$

We are given:

$$\begin{aligned} P &= 16x^3 - 6x^2 + 12x + 4 \\ \text{Side}_1 &= 8x^2 + 3x \end{aligned}$$

First, compute half of the perimeter:

$$\frac{P}{2} = \frac{16x^3 - 6x^2 + 12x + 4}{2} = 8x^3 - 3x^2 + 6x + 2$$

Now, subtract the given side (Side_1) from half the perimeter to obtain the other side (Side_2):

$$\begin{aligned} \text{Side}_2 &= (8x^3 - 3x^2 + 6x + 2) - (8x^2 + 3x) \\ &= 8x^3 - 3x^2 + 6x + 2 - 8x^2 - 3x \\ &= 8x^3 + (-3x^2 - 8x^2) + (6x - 3x) + 2 \\ &= 8x^3 - 11x^2 + 3x + 2 \end{aligned}$$

Answer: The other side of the rectangle is $8x^3 - 11x^2 + 3x + 2$ units.